DA5500W — Data Mining: Concepts and Techniques

Tier	B
Phase (order)	5
Do this first	DA2402, DA2400
Estimated effort	10 weeks
Unlocks	DA3406

Roadmap

1. Scraping, descriptive stats, pattern mining, clustering, visualisation.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Data Mining for Decision Making (35 videos). NPTEL-NOC IITM *Data Mining for Decision Making*.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Data Mining (36 videos). NPTEL *Data Mining* (Pabitra Mitra, IIT Kharagpur, re-upload): follows Han & Kamber.
- Stanford CS224W Machine Learning with Graphs I Jure Leskovec (47 videos). Stanford CS224W *Machine Learning with Graphs* (Jure Leskovec, official): link analysis, graph mining.

⚡ Short videos for intuition and revision

- Data Warehouse & Data Mining (25 videos). Gate Smashers *Data Warehouse & Data Mining*: short revision.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–4	What data mining is, decision lists, association rules; install WEKA (you can do every exercise in Python/scikit-learn instead).
2	5–10	Naive Bayes (categorical and numeric), decision trees. Run both on the UCI Adult dataset.
3	11–18	Neural networks, clustering, PCA.

Week	Lectures	What to do
4	19–29	Linear and logistic regression, kNN, market-basket analysis (Apriori), LDA. Mine association rules from a groceries dataset.
5	30–35	Evaluation metrics, SVMs. Then the free book *Mining of Massive Datasets* (mmds.org) for scale: LSH, streams, PageRank.

▼ **All lectures in *Data Mining for Decision Making* (35 videos)**

1. Data Mining for Decision Making | INTRO | NPTEL | 2026
2. Data Mining Explained: Decision Lists & Association Rules Made Simple
3. How to Install WEKA: Step-by-Step Guide for Beginners
4. Getting Started with WEKA: A Complete Beginner's Overview
5. Basics of Bayesian Classification: Understanding Probabilistic Learning
6. A Deep Dive into Bayesian Classification: Concepts & Intuition building
7. Implementing Bayesian Classification Using WEKA
8. Bayesian Classification with Numeric Data: Theory & Examples
9. Decision Trees Explained: From Roots to Leaves
10. Building Decision Trees in WEKA: Hands-On Tutorial
11. Neural Networks Fundamentals: How Artificial Brains Learn
12. Neural Networks Explained: Architecture & Learning
13. Implementing Neural Networks in WEKA
14. Clustering Basics: Grouping Data Without Labels
15. Clustering Algorithms in WEKA: Practical Demonstration
16. Clustering Techniques Explained: Beyond the Basics
17. Principal Component Analysis (PCA): Dimensionality Reduction Explained
18. Applying PCA in WEKA: Step-by-Step Walkthrough
19. Linear Regression Explained: Modeling Relationships in Data
20. Linear Regression Using WEKA: Practical Implementation
21. Logistic Regression Basics: Classification Made Easy
22. Logistic Regression in WEKA: Hands-On Classification
23. Instance-Based Learning Explained: Learning from Examples
24. Instance-Based Learning in WEKA: KNN and Beyond
25. Market Basket Analysis Explained: Discovering Hidden Patterns
26. Market Basket Analysis Using WEKA
27. Linear Discriminant Analysis (LDA): Classification Intuition
28. Linear Discriminant Analysis Explained in Depth
29. Implementing LDA in WEKA
30. Model Evaluation Metrics Explained: Accuracy, Precision & More
31. Support Vector Machines Explained: Maximum Margin Learning
32. Support Vector Machines Deep Dive: Kernels & Optimization
33. Advanced SVM Concepts: Kernels, Margins & Hyperparameters

34. Support Vector Machines in WEKA: Practical Tutorial
35. Data Mining for Decision Making: Key Takeaways & Final Insights

Books, notes and websites

- Book: Han, Kamber, Pei; Tan et al.; Mining of Massive Datasets (mmds.org)

Practice / proof of understanding

Scrape a site and mine frequent patterns.

What your IITM course sheet says

From the PDF: DA5500W

Official description: This course prepares students to apply data mining and visualization techniques to real-world problems across diverse domains. By the end of the course, students will be able to: ? Understand fundamental concepts and methods of data mining. ? Acquire skills in data collection, preprocessing, visualization, and mining. ? Apply clustering, outlier detection, and advanced visualization techniques. ? Gain practical experience through scraping, mining, and visualization assignments.

Official syllabus (trimmed):

> Module 1: Introduction to Data Mining and Data Types o KDD process, scope, applications o Issues in data mining o Data types: relational, transactional, time-series, text, graph, multimedia Module 2: Data Scraping & Collection o Data acquisition: APIs, scraping, open datasets o Tools: BeautifulSoup, Scrapy, Selenium o Ethical & legal considerations Module 3: Understanding Data: Attributes & Basic Statistics with Visualization o Attribute types (nominal, ordinal, numeric, binary) o Descriptive statistics (mean, variance, quartiles, five-number summary) o Visualization: histograms, scatter plots, q-q plots o Similarity & dissimilarity measures Module 4: Data Preprocessing – Cleaning & Integration o Data quality dimensions o Handling missing values, noisy data, outliers o Integration: redundancy handling, conflict resolution Module 5: Data Preprocessing – Reduction & Transformation o Dimensionality reduction: PCA, feature selection o Numerosity reduction: sampling, regression, histograms o Normalization & discretization Module 6: Advanced Data Visualization & Visual Data Mining o Pixel-oriented, geometric projection (parallel coords, t-SNE/UMAP) o Hierarchical visualizations (treemaps, sunburst, radial trees) o Icon-based, graph/network visualization o Text visualization (tag clouds, topic maps) o Visualizing results & mining process pipelines o Interact

Official textbooks: 1. Han, J., Kamber, M., & Pei, J. (2012). Data Mining: Concepts and Techniques. 3rd Ed. Morgan Kaufmann. 2. Tan, P.-N., Steinbach, M., & Kumar, V. (2018). Introduction to Data Mining. Pearson.

Official references: 1. Aggarwal, C. C. (2015). Data Mining: The Textbook. Springer. 2. Munzner, T. (2014). Visualization Analysis and Design. CRC Press. 3. Mitchell, R. (2018). Web Scraping with Python. O’Reilly Media.